

Physics Playbook

Matter · Measurement · Force · Simple Machines · Energy · Light · Magnetism

ICSE Class 6 · Full Book · Final Exam Ready

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Chapter names and order can differ slightly between schools and publishers. Follow your school's exam syllabus.

How to use this playbook

Each chapter has notes, a 🧠 memory trick, a 🎲 activity you can do in class with simple things, a 📍 Mumbai connection and a ✏️ quick check. The answers are at the end.

Chapter	Topic
1	Matter
2	Physical Quantities and Measurement
3	Force
4	Simple Machines
5	Energy
6	Light
7	Magnetism

⭐ **Exam Tip: How to score full marks in Physics**

1. Learn the **definitions word for word**. They are asked every year.
2. Always write the **SI unit** with every numerical answer.
3. Draw **labelled diagrams** with a pencil and ruler.
4. For "give a reason" questions, use the pattern **"because ..., therefore ..."**.

1. Matter

- **Matter** is anything that has **mass** and occupies **space** (volume). Air, water, a stone and our body are all matter.
- Light, heat and sound are **not** matter. They have no mass and take up no space. They are forms of energy.
- Matter is made of tiny particles called **molecules**, and molecules are made of **atoms**.

Properties of the particles (molecules)

1. They are very **small**.
2. There are **spaces** between them (**intermolecular space**).
3. They **attract** each other (**intermolecular force**).
4. They are always **moving**.

Three states of matter

Property	Solid	Liquid	Gas
Shape	Fixed	Not fixed (takes the shape of its container)	Not fixed
Volume	Fixed	Fixed	Not fixed (fills its whole container)
Intermolecular space	Very small	Larger	Largest
Intermolecular force	Strongest	Weaker	Weakest
Movement of particles	Only vibrate about fixed positions	Move freely within the liquid	Move freely and fast in all directions
Compressibility	Almost none	Very little	Highly compressible
Example	Ice, wood	Water, milk	Air, steam

Change of state

Change	Name	Example
Solid → liquid	Melting (at the **melting point**)	Ice → water at 0 °C
Liquid → solid	Freezing	Water → ice
Liquid → gas	Vaporisation or boiling (at the **boiling point**)	Water → steam at 100 °C
Gas → liquid	Condensation	Water drops on a cold bottle
Solid → gas directly	Sublimation	Camphor, naphthalene balls, dry ice

- **Evaporation** happens at **any temperature** and only from the **surface**. Boiling happens at a **fixed temperature** and throughout the liquid.
- Heating makes the particles move faster and further apart. Cooling does the opposite.

🧠 Memory Trick

Solids are like students in an exam hall: fixed seats, only fidgeting (vibrating). **Liquids are like students in the corridor**: close together but moving around. **Gases are like students when the bell rings for recess**: running everywhere, far apart!

📍 Mumbai Connection

In the Mumbai monsoon, clothes take long to dry because the air is already full of water vapour (humid), so **evaporation is slow**. A cold drink bottle "sweats" because the water vapour in the air **condenses** on it.

🎲 Play It in Class: Be the Molecules

Students act as molecules. "Solid!" means they stand close, holding hands and swaying. "Liquid!" means they let go but stay close and slide past each other. "Gas!" means they spread across the room and move fast. The teacher calls out "Heat it!" or "Cool it!" to change the state.

✏️ Quick Check

1. Define matter.
2. Why can a gas be compressed easily, but a solid cannot?
3. Name the change: (a) naphthalene balls getting smaller (b) dew forming on grass.
4. Give two differences between evaporation and boiling.

2. Physical Quantities and Measurement

- A **physical quantity** is anything that can be measured: length, mass, time, area, volume, speed, temperature.
- **Measurement** = a **number** + a **unit**. For example, 5 m: "5" is the number and "m" is the unit.
- **SI system** (International System of Units) is the one standard set of units used by everyone.

SI units

Quantity	SI unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Area	square metre	m^2
Volume	cubic metre	m^3
Speed	metre per second	m/s

Length

- 1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm
- **Measuring with a ruler:** place the object at the 0 mark. If the end is broken, start from 1 cm and subtract 1. Keep your eye **directly above** the reading to avoid the **parallax error**.
- **Curved lines:** use a **thread**, then measure the thread on a ruler. A **divider** also works on maps.

Mass and weight

Mass	Weight
The **quantity of matter** in a body	The **force** with which the Earth pulls a body
Stays the same everywhere	Changes from place to place (less on the Moon)
Measured with a **beam balance**	Measured with a **spring balance**
SI unit: kg	SI unit: newton (N)

- 1 kg = 1000 g, 1 tonne = 1000 kg, 1 quintal = 100 kg

Time

- 1 minute = 60 s, 1 hour = 3600 s, 1 day = 86,400 s
- **Instruments:** sundial (ancient), hourglass (sand clock), water clock, pendulum clock, quartz clock, stopwatch
- A **simple pendulum** swings with a fixed time period, which is why it can be used to measure time.

Area and volume

- **Area of a regular shape:** length × breadth. **Irregular shape:** use **graph paper** and count the squares (a full square = 1, more than half = 1, less than half = 0).
- **Volume of a regular solid:** length × breadth × height
- **Volume of an irregular solid** (such as a stone): use a **measuring cylinder** and the **displacement of water**. Volume = final level − initial level.
- **Volume of liquids:** 1 litre (L) = 1000 millilitres (mL), and **1 mL = 1 cm^3**
- Read the **bottom of the curved surface** (meniscus) of water, with your eye level with it.

🧠 Memory Trick

The SI starter pack: "Meter, Kilo, Second": m, kg, s, "My Kite Soars". **Mass never changes, weight goes on a diet on the Moon!** (weight on the Moon = 1/6 of weight on Earth)

📍 Mumbai Connection

Mumbai to Pune is about 150 km = 1,50,000 m. A Mumbai local train is about 12 coaches long, roughly 250 m. Ask students: "How many coaches would fit between Dadar and Bandra (about 5 km)?"

🎲 Play It in Class: Measure Without a Ruler

Teams measure the classroom using hand spans, foot lengths and pencils, then compare the answers. Why are they different? That's exactly why we need **standard units**!

✏️ Quick Check

1. What is the SI unit of (a) mass (b) time (c) volume?
2. Convert: (a) 2.5 km into m (b) 3 hours into seconds.
3. Give two differences between mass and weight.
4. A measuring cylinder reads 40 mL. After a stone is dropped in, it reads 55 mL. Find the volume of the stone.

3. Force

- **Force** is a **push or a pull** on an object. Its SI unit is the **newton (N)**.

What force can do (effects of force)

1. Move a stationary object (kicking a ball)
2. Stop a moving object (catching a ball)
3. Change speed (pedalling faster)
4. Change direction (hitting a cricket ball with a bat)
5. Change shape or size (pressing a sponge or dough)

Types of force

Contact forces (need touching)	Non-contact forces (act from a distance)
Muscular force: lifting a bag	**Gravitational force**: the Earth pulls everything down
Frictional force: opposes motion between surfaces	**Magnetic force**: a magnet pulls iron
Normal or reaction force	**Electrostatic force**: a rubbed comb attracts bits of paper

Friction

- **Friction** is the force that **opposes motion** between two surfaces in contact.
- **Rough surfaces → more friction. Smooth surfaces → less friction.**
- **Useful:** helps us walk, write, hold things and brake vehicles.
- **Harmful:** wears out shoes and tyres, produces heat in machines, wastes energy.
- **Increasing friction:** treads on tyres, grooves on shoe soles, sand on slippery roads
- **Reducing friction:** oil or grease (lubricants), ball bearings, wheels, polishing, streamlined shapes

Pressure (basic idea)

- **Pressure = Force ÷ Area.** The same force on a **smaller area** gives **more pressure**.
- A sharp knife cuts easily (small area). A school bag with **wide straps** hurts less (large area).

> 🧠 **Memory Trick**
> **The 5 effects of force: "Move, Stop, Speed, Turn, Shape" (MS STS). Friction is the "naughty brake"**: it always fights against motion.

> 📍 **Mumbai Connection**
> Monsoon roads in Mumbai are slippery because water **reduces friction** between tyres and the road. That's why bikes skid, and why tyres have deep **treads**.

🎲 Play It in Class: Friction Race

Slide the same book down a tilted board three times: on bare wood, on a towel and on cling film. Which surface is fastest? Students rank the surfaces from most to least friction.

✏️ Quick Check

1. Define force and give its SI unit.
2. Name the force: (a) an apple falling from a tree (b) a comb attracting paper bits.
3. Give two ways to reduce friction.
4. Why do school bags have broad straps?

4. Simple Machines

- A **machine** is a device that makes work **easier**: it lets a small force (**effort**) overcome a large force (**load**), changes the **direction** of a force, or increases **speed**.
- **Load:** the weight lifted. **Effort:** the force applied. **Fulcrum:** the fixed point a lever turns about.

The lever

Principle of the lever: Load × load arm = Effort × effort arm

Class	What is in the middle	Examples
Class I	**Fulcrum** (F in the middle)	See-saw, scissors, pliers, beam balance, crowbar
Class II	**Load** (L in the middle)	Nutcracker, bottle opener, wheelbarrow, lemon squeezer
Class III	**Effort** (E in the middle)	Tweezers (forceps), fishing rod, broom, human forearm, sugar tongs

- Class II levers always **save effort**. Class III levers **don't save effort**, but they increase speed and convenience.

Other simple machines

Machine	How it helps	Example
Pulley	Changes the direction of the force (pull down to lift up)	A well, a flagpole
Inclined plane	Less effort to lift a load up a slope	A ramp, a slide, a staircase
Wedge	Splits or cuts things	An axe, a knife, a nail
Screw	An inclined plane wrapped around a rod	A jar lid, a bolt
Wheel and axle	A big wheel turns a small axle with little effort	A steering wheel, a doorknob, a screwdriver

- **Mechanical advantage (MA)** = Load ÷ Effort. If MA > 1, the machine saves effort.
- Machines need **oiling** to reduce friction, so that less energy is wasted.

> 🧠 **Memory Trick**
> **"F-L-E = 1-2-3"**: whatever is in the **middle** decides the class. Class **1** → **F**ulcrum in the middle. Class **2** → **L**oad in the middle. Class **3** → **E**ffort in the middle.

> 📍 **Mumbai Connection**
> The ramps at Mumbai railway stations and footbridges are **inclined planes**. They let people push trolleys and wheelchairs up with less effort.

🎲 Play It in Class: Lever Hunt

Bring scissors, a nutcracker, tweezers, a bottle opener and a ruler balanced on an eraser. Teams find F, L and E on each tool and sort the tools into Class I, II and III. The quickest correct team wins.

✏️ Quick Check

1. Define fulcrum, load and effort.
2. State the principle of the lever.
3. Name the class of lever: (a) a see-saw (b) a nutcracker (c) a fishing rod.
4. A load of 200 N is lifted with an effort of 50 N. Find the mechanical advantage.

5. Energy

- **Energy** is the **capacity to do work**. Its SI unit is the **joule (J)**.
- **Work** is done when a force moves an object. Work = Force × distance moved.

Forms of energy

Form	Example
Mechanical: **kinetic** (from motion)	A moving car, flowing water, wind
Mechanical: **potential** (from position or shape)	Water stored in a dam, a stretched bow, a wound spring
Heat	A hot iron, the Sun
Light	A bulb, the Sun
Sound	A bell, music
Chemical	Food, fuel, a battery
Electrical	Current in wires
Nuclear	The Sun and nuclear power plants

Energy changes (conversion)

Device	Energy change
Electric bulb	Electrical → light + heat
Electric fan	Electrical → mechanical (kinetic)
Battery or cell	Chemical → electrical
Solar cell	Light → electrical
Microphone	Sound → electrical
Loudspeaker	Electrical → sound
Our body	Chemical (food) → mechanical + heat

- **Law of conservation of energy:** energy can **neither be created nor destroyed**. It only changes from one form to another.

Sources of energy

Renewable (will not run out)	Non-renewable (will run out)
Solar, wind, water (hydro), tides, biomass	Coal, petroleum, natural gas (fossil fuels)

- **Save energy:** switch off fans and lights, use LED bulbs, use public transport, use solar energy.

> 🧠 **Memory Trick**
> **Kinetic = "Kicking" (moving). Potential = "Pending" (stored, waiting to be used). Energy is like money in a bank: you can change ₹ into $, but the value is never lost!**

📍 Mumbai Connection

Mumbai's local trains run on **electrical** energy and turn it into **kinetic** energy. Many Mumbai buildings now have **solar panels** on their roofs, which turn light energy into electrical energy.

🎲 Play It in Class: Energy Chain

Start with "Sun" and pass a ball around. Each student adds the next step: Sun → light → plant (chemical) → we eat the food (chemical) → we cycle (kinetic) → the dynamo lights a bulb (electrical → light). Whoever breaks the chain starts the next one.

✏️ Quick Check

1. Define energy and give its SI unit.
2. Name the energy change in (a) a loudspeaker (b) a solar cell.
3. Give two examples of potential energy.
4. State the law of conservation of energy.

6. Light

- **Light** is a form of energy that makes things **visible**.
- **Luminous objects** give out their own light (the Sun, a bulb, a candle). **Non-luminous objects** don't (the Moon, a book). We see them by **reflected** light.
- **Light travels in a straight line** (**rectilinear propagation**). Proof: a pinhole camera, or looking through straight and bent pipes.
- Speed of light = **3×10^8 m/s** (the fastest thing known).

Materials and light

Type	Light passes through?	Example
Transparent	Fully	Clear glass, water, air
Translucent	Partly	Butter paper, frosted glass, oiled paper
Opaque	Not at all	Wood, metal, a book

Shadows

- A **shadow** forms when an **opaque** object blocks light.
- A shadow is always **dark**, it forms on the **opposite side** from the light source, and it needs a **screen** (the ground or a wall).
- **Umbra:** the dark central part. **Penumbra:** the lighter outer part.
- **Eclipses** are shadows in space. In a **solar eclipse** the Moon comes between the Sun and the Earth. In a **lunar eclipse** the Earth comes between the Sun and the Moon.

Reflection

- **Reflection** is light bouncing back from a shiny surface such as a **mirror**.
- **Incident ray:** comes in. **Reflected ray:** goes out. **Normal:** a line at 90° to the mirror.
- **Laws of reflection:** (1) the angle of incidence = the angle of reflection ($\angle i = \angle r$) (2) the incident ray, the reflected ray and the normal all lie in the **same plane**.
- **The image in a plane mirror:** upright, the same size, as far behind the mirror as the object is in front, virtual, and **laterally inverted** (left and right are swapped).

Pinhole camera

- Its image is **inverted** (upside down) and **real**. This proves that light travels in straight lines.

> 🧠 **Memory Trick**
> **"I = R" (eye equals our)**: the angle of **I**ncidence = the angle of **R**eflection. **"AMBULANCE" is written backwards on an ambulance** so drivers can read it in their mirror, because of **lateral inversion**!

📍 Mumbai Connection

At Marine Drive in the evening, the long shadows of buildings stretch across the road, because the low Sun makes **long shadows**. At noon, shadows are **shortest**.

🎲 Play It in Class: Shadow Puppet Show

Use a torch and a white wall. Students make animal shadows with their hands. Ask: "What happens when the hand moves closer to the torch?" (the shadow gets bigger). Then try it with glass, butter paper and a book to classify materials as transparent, translucent or opaque.

✏️ Quick Check

1. What are luminous objects? Give two examples.
2. Classify as transparent, translucent or opaque: (a) butter paper (b) a steel plate (c) clear water.
3. State the two laws of reflection.
4. Why is "AMBULANCE" written in reverse on ambulances?

7. Magnetism

- A **magnet** attracts **magnetic materials** (iron, nickel, cobalt, steel). Non-magnetic materials (wood, plastic, paper, copper, aluminium) are not attracted.
- **Natural magnet: lodestone** (magnetite), discovered long ago. **Artificial magnets:** bar, horseshoe, U-shaped, ring, cylindrical and needle magnets.

Properties of magnets

1. A magnet **attracts** iron, nickel and cobalt.
2. The attraction is **strongest at the poles** (the two ends).
3. A freely hanging magnet always points **north–south** (the **directive property**).
4. **Like poles repel and unlike poles attract** (N–N repel, N–S attract).
5. **Poles always exist in pairs.** Breaking a magnet gives two smaller magnets, each with its own N and S poles.

- **Repulsion is the surest test of a magnet**, since a magnet can attract plain iron too.

Making a magnet

- **Single touch method:** stroke an iron nail with one pole of a magnet, **in one direction only**, about 30–40 times.
- **Electrical method:** wrap wire around an iron nail and pass current through it to make an **electromagnet**. It is magnetic only while the current flows.

Losing magnetism

- Heating, hammering, dropping again and again, and storing magnets badly all weaken them.
- **Safe storage:** keep bar magnets in **pairs, with unlike poles next to each other**, and soft iron pieces (**keepers**) across the ends.

Uses

- **Magnetic compass** (for direction), electric bells, speakers, MRI machines, cranes in scrapyards, fridge doors, maglev trains

> 🧠 **Memory Trick**
> **"Opposites attract, sames are shy"**: N and S attract, N and N repel. **Magnetic metals: "I Never Cry"**: **I**ron, **N**ickel, **C**obalt.

> 📍 **Mumbai Connection**
> Ships entering **Mumbai Port** and fishing boats going out into the Arabian Sea carry a **compass** to find their direction. It works because of the magnet's **directive property**.

🎲 Play It in Class: Magnet Treasure Hunt

Mix paper clips, coins, plastic buttons, aluminium foil, safety pins and erasers in a bowl. Teams predict which items a magnet will pick up, then test them. Count the correct predictions. (Surprise: some Indian coins **are** magnetic and some are not. Test them and see!)

✏️ Quick Check

1. Name three magnetic materials.
2. What is the directive property of a magnet?
3. Why is repulsion the surest test of a magnet?
4. How should bar magnets be stored?

Full-Book Revision Test

Section A (1 mark each)

1. What is the SI unit of force?
2. Name a substance that sublimes.
3. Give one example of a Class II lever.
4. Name the energy change in an electric bell.
5. What kind of material is frosted glass?
6. Name the natural magnet.

Section B (2–3 marks each)

7. Give three differences between solids and gases.
8. Describe how to find the volume of an irregular stone.
9. Give two advantages and two disadvantages of friction.
10. Draw and label a Class I lever. Give two examples.
11. Distinguish between renewable and non-renewable sources of energy, with two examples of each.
12. List the characteristics of the image formed by a plane mirror.

Section C (4 marks each)

13. Explain why a shadow is formed, and name the two parts of a shadow.
14. A load of 300 N is placed 20 cm from the fulcrum. What effort is needed at 60 cm from the fulcrum to balance it?
15. Explain the single touch method of making a magnet.

Answer Key

Ch 1: (1) Matter is anything that has mass and occupies space. (2) A gas has large spaces between its particles, so they can be pushed closer. A solid's particles are already very close. (3) (a) Sublimation (b) Condensation (4) Evaporation happens at any temperature, only from the surface. Boiling happens at a fixed temperature, throughout the liquid.

Ch 2: (1) (a) kg (b) s (c) m^3 (2) (a) 2500 m (b) 10,800 s (3) Mass is the quantity of matter and doesn't change from place to place (kg). Weight is the pull of gravity and does change (N). (4) 15 mL = 15 cm^3

Ch 3: (1) A push or a pull. Its SI unit is the newton (N). (2) (a) Gravitational force (b) Electrostatic force (3) Lubricants (oil), ball bearings, polishing (any two) (4) Broad straps increase the area, so the pressure on the shoulders is less.

Ch 4: (1) Fulcrum: the fixed point. Load: the weight lifted. Effort: the force applied. (2) Load × load arm = Effort × effort arm (3) (a) Class I (b) Class II (c) Class III (4) MA = 200 ÷ 50 = 4

Ch 5: (1) The capacity to do work. Its SI unit is the joule (J). (2) (a) Electrical → sound (b) Light → electrical (3) Water stored in a dam, a stretched rubber band or bow, a wound spring (4) Energy can neither be created nor destroyed. It only changes from one form to another.

Ch 6: (1) Objects that give out their own light: the Sun, a candle, a bulb. (2) (a) Translucent (b) Opaque (c) Transparent (3) ∠i = ∠r, and the incident ray, reflected ray and normal lie in the same plane. (4) Because of lateral inversion, drivers ahead read it correctly in their rear-view mirrors.

Ch 7: (1) Iron, nickel, cobalt (2) A freely hanging magnet always comes to rest pointing north–south. (3) Plain iron is attracted by both poles, but only a magnet can repel another magnet. (4) In pairs with unlike poles side by side, with keepers (soft iron pieces) across the ends.

Revision test: (1) Newton (2) Camphor, naphthalene or dry ice (3) Nutcracker, wheelbarrow or bottle opener (4) Electrical → sound (5) Translucent (6) Lodestone (magnetite) (7) Shape (fixed / not fixed), volume (fixed / not fixed), compressibility (none / high), intermolecular space (small / large) (8) Fill a measuring cylinder partly with water and note the level. Gently drop in the stone tied to a thread and note the new level. The volume = the difference. (9) Advantages: we can walk, brakes work. Disadvantages: wear and tear, heat and wasted energy. (10) The fulcrum is between the load and the effort. Examples: see-saw, scissors. (11) Renewable sources can be replaced naturally (solar, wind). Non-renewable sources will run out (coal, petroleum). (12) Upright, the same size, virtual, laterally inverted, and as far behind the mirror as the object is in front. (13) An opaque object blocks the straight-line path of light, so a dark area forms behind it. The two parts are the umbra and the penumbra. (14) 300 × 20 = E × 60, so E = **100 N** (15) Place the iron bar on a table. Stroke it from one end to the other with one pole of a magnet. Lift the magnet high at the end and repeat in the same direction 30–40 times. The end where the stroke finishes gets the opposite pole.